

Jastrow-type Decomposition in Quantum Chemistry for Low-Depth Quantum Circuits

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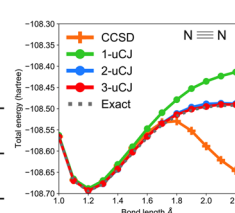
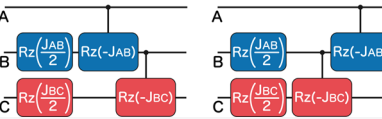
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ABSTRACT: We propose an efficient $O(N^2)$ -parameter ansatz that consists of a sequence of exponential operators, each of which is a unitary variant of Neuscamman's cluster Jastrow operator. The ansatz can also be derived as a decomposition of T_2 amplitudes of the unitary coupled cluster with generalized singles and doubles, which gives a near full-CI energy. The proposed ansatz therefore can reproduce the uCCGSD energy, or rather will reach the exact full-CI energy because of the exponential operator product form. Because the cluster Jastrow operators are expressed by a product of number operators and the derived Pauli operator products, namely, the Jordan–Wigner strings, are all commutative, it does not require the Trotter approximation to implement to a quantum circuit and should be a good candidate for the variational quantum eigensolver algorithm of a near-term quantum computer. The accuracy of the ansatz was examined for dissociation of a nitrogen dimer, and compared with other existing $O(N^2)$ -parameter ansatzs. Not only the original ansatzs defined in the second-quantization form but also their Trotter-splitting variants, in which the cluster amplitudes are optimized to minimize the energy obtained with a few, typically single, Trotter steps, were examined by quantum circuit simulators.

Trotter-free CC ansatz

$$\hat{T}_{\text{uCC}} = e^{\hat{K}} e^{\sum J_{\alpha\beta\gamma\delta} f_{\alpha\beta\gamma\delta}} e^{\hat{K}'} e^{\sum J_{\alpha\beta\gamma\delta} f_{\alpha\beta\gamma\delta}} \dots$$

$$\rightarrow e^{J_{\alpha\beta\gamma\delta} f_{\alpha\beta\gamma\delta}} e^{J_{\alpha\beta\gamma\delta} f_{\alpha\beta\gamma\delta}} \dots = e^{J_{\alpha\beta\gamma\delta} f_{\alpha\beta\gamma\delta}} e^{J_{\alpha\beta\gamma\delta} f_{\alpha\beta\gamma\delta}} \dots$$



1. INTRODUCTION

Quantum chemistry is gaining more attention as a promising field of application of quantum computers because of its high affinity to the qubit representations and universal gate operations.^{1–5} An obvious advantage of quantum computers on quantum chemical calculations is that occupation-number vectors in the second quantization formulation can be naturally mapped to the qubit representation by the Jordan–Wigner or Bravyi–Kitaev transformations, and arbitrary state in the vector space and unitary transformation are expressed by the qubits and quantum gates, while the space is usually truncated for the case of classical computers because the dimension of the vector space, namely, the full-CI space, grows exponentially with the number of electrons.

While quantum algorithms for numerical linear algebra is currently being developed,^{6–8} so far there are two main approaches for computing eigenvalues of the second-quantized electronic Hamiltonian. The first developed method is the phase estimation algorithm (PEA),^{9,10} in which an approximated eigenstate prepared on quantum registers is propagated with the Hamiltonian to encode the eigenvalue into the relative phases of binary states of ancilla qubits, and the phase is decoded by a fractional binary expansion via the inverse quantum Fourier transformation (QFT). In theory, a desired precision ϵ can be achieved at the cost of $O(1/\epsilon)$; however, in reality, it largely increases the demand for coherence time, which is too severe for near-term quantum computers without

error corrections. There have been many attempts to mitigate the requirements, such as the iterative PEA,^{11,12} which does not require the QFT. The second method is the variational quantum eigensolvers (VQEs)^{13,14} in which quantum computers are used only for evaluating energy expectation values of trial wave functions that are defined by a set of parameters, and classical computers are used for updating the parameters. One of the advantages of the VQE over the PEA is that once a trial wave function is prepared, the measurements only require simple single qubit rotations, which should greatly reduce the coherence time requirements, ideally $O(1)$.¹⁵ In addition, the recent developments of the quantum algorithms^{16–19} for the evaluation of the derivatives with respect to the parameters in the exponents should greatly accelerate the variational optimization. The VQE is therefore a promising candidate algorithm in the noisy intermediate scale quantum (NISQ) computers era,²⁰ although the required times of measurements increases as $O(1/\epsilon^2)$ to achieve a desired precision ϵ .

It is therefore important to develop an efficient wave function ansatz that requires only a small number of quantum gates to prepare on a quantum circuit. The widely used unitary

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coupled cluster with singles and doubles (uCCSD) ansatz^{21–23} has $O(N^4)$ variational parameters and the depth should grow as $O(N^3)$ at least. A number of $O(N^2)$ -parameter ansatz have been developed, such as the low-depth ansatz,²⁴ and low-rank approximation,²⁵ and recently, k -fold products of unitary pair coupled cluster with generalized singles and doubles (UpCCGSD).²⁶ Another obstacle to implement exponential operator ansatz to quantum circuits is that all the Pauli operator products produced by the Jordan–Wigner or Bravyi–Kitaev transformations of the second quantized operators in an exponent need to be split to a product of the exponential operators, each of which has only a single Pauli product in the exponent. It means that many Trotter steps should be performed before the measurement if one wants to compute the energy expectation value accurately as provided by the definition written in the second quantized operators because they are usually not commutative. One way to work around this problem is to adopt hardware-oriented ansatzs, such as the hardware-efficient ansatz²⁷ or an incremental adaptive scheme.^{28,29}

In this study, we present an efficient $O(N^2)$ -parameter ansatz for the VQE algorithm termed k -uCJ that consists of repeated k -times multiplication of a unitary variant of the cluster Jastrow (CJ) exponential operator^{30,31} and can reproduce the generalized unitary coupled cluster (uCCGSD) ansatz,^{26,32,33} which can give a near full-CI energy, or rather will reach the exact full-CI energy by increasing k .

2. THEORY

In this section, matrices can be real or complex. For clarification, they are denoted by italic and calligraphy letters, e.g., K and $\mathcal{K}$, respectively.

2.1. Unitarization of the CJ Ansatz. An efficient parametrization for the T_2 amplitudes of generalized CC, $T_{p_o q_o r_s r_t}^{\text{CJ}} \simeq T_{p_o q_o r_s r_t}^{\text{CJ}} = \sum_{jl} U_{pj} U_{ql} J_{jl}^{(\sigma\tau)} U_{rl} U_{sl}$ named as cluster Jastrow (CJ) ansatz, was introduced by Neuscamman^{30,31} in his cluster Jastrow anti-symmetric geminal power (CJ-AGP) ansatz. Simultaneous optimization of the parameters U and J and the AGP reference function with a variational Monte Carlo method provided an efficient and accurate description of both weak and strong electron correlations.^{30,31,34,35}

In the CJ ansatz, the 4-index cluster operator $\hat{T}^{\text{CJ}}$ is decomposed into two 2-index operators

$$e^{\hat{T}^{\text{CJ}}} = e^{-\hat{K}} e^{\hat{J}} e^{\hat{K}} \quad (1)$$

where

$$\hat{T}^{\text{CJ}} = \sum_{pqrs, \sigma\tau} T_{p_o q_o r_s r_t}^{\text{CJ}} a_{p_o}^\dagger a_{q_o}^\dagger a_{r_s} a_{r_t} \quad (2)$$

$$\hat{J} = \sum_{jl, \sigma\tau} J_{jl}^{\sigma\tau} a_{j_o}^\dagger a_{j_s}^\dagger a_{l_s} a_{l_t} \quad (3)$$

$$\hat{K} = \sum_{p < q, \sigma} K_{pq} (a_{p_o}^\dagger a_{q_o} - a_{q_s}^\dagger a_{p_s}) \quad (4)$$

while $e^{\hat{K}}$ is an orbital rotation operator and unitary $\hat{J}$ is symmetric with respect to permutation of the indices j and l , thus $e^{\hat{J}}$ cannot be unitary if one restricts J to real matrices. The matrix $J^{\sigma\tau}$ needs to be replaced by pure imaginary matrices $\mathcal{J}$ to make $e^{\hat{J}}$ a unitary operator, and we introduced a unitary

variant of the CJ ansatz, uCJ ansatz, by replacing $\hat{K}$ and $\hat{J}$ in eq 1 by

$$\hat{K} = \sum_{pq, \sigma} \mathcal{K}_{pq} a_{p_o}^\dagger a_{q_o} \quad (5)$$

$$\hat{J} = \sum_{jl, \sigma\tau} \mathcal{J}_{jl}^{(\sigma\tau)} a_{j_o}^\dagger a_{j_s}^\dagger a_{l_s} a_{l_t} \quad (6)$$

where $\mathcal{K}$ is a complex anti-Hermit matrix and $\mathcal{J}^{\alpha\alpha} (= \mathcal{J}^{\beta\beta})$ and $\mathcal{J}^{\alpha\beta} (= \mathcal{J}^{\beta\alpha})$ are pure imaginary symmetric matrices with respect to the indices j and l , respectively. Note that when we restricted $\mathcal{K}$ to real anti-symmetric matrices and optimized it simultaneously with $\mathcal{J}$, energy expectation values by the ansatz never fell below those of reference wave functions. The cluster amplitudes of uCJ are therefore generally complex numbers

$$T_{p_o q_o r_s r_t} \simeq \mathcal{T}_{p_o q_o r_s r_t}^{\text{uCJ}} = \sum_{jl} \mathcal{U}_{pj} \mathcal{U}_{ql}^* \mathcal{J}_{jl}^{(\sigma\tau)} \mathcal{U}_{rl} \mathcal{U}_{sl}^* \quad (7)$$

It is possible to restrict the amplitudes to real numbers by adding its complex conjugate $\mathcal{T}_{p_o q_o r_s r_t}^{\text{uCJ}} + \mathcal{T}_{p_o q_o r_s r_t}^{\text{uCJ}*}$ but we found that the variational energy is not different from that with eq 7.

It would be beneficial to see the other choice of orbital pairing for the decomposition of the T_2 amplitudes other than eq 7. If one decomposes the amplitudes as $U_{pj} U_{ql} J_{jl}^{(\sigma\tau)} U_{rl} U_{sl}$ i.e., (p, r) and (q, s) are respectively paired, the transformed quadratic operator in eq 3 becomes $\hat{J}' = \sum_{jl, \sigma\tau} J_{jl}^{(\sigma\tau)} a_{j_o}^\dagger a_{l_o}^\dagger a_{r_s} a_{r_t}$, which, in fact, corresponds to the cluster operator used in the UpCCGSD ansatz. A comparison of the performance between the uCJ and UpCCGSD ansatzs will be presented in Section 3.

2.2. k -Fold Products of uCJ Ansatz. To recover the uCCGSD limit in eq 7, it is natural to extend the uCJ ansatz by introducing an extra index x to $\mathcal{J}$ and $\mathcal{K}$ as

$$\mathcal{T}_{p_o q_o r_s r_t} \simeq \sum_{x=1}^k \sum_{jl} \mathcal{U}_{pj}^x \mathcal{U}_{ql}^{x*} \mathcal{J}_{jl}^{(\sigma\tau)x} \mathcal{U}_{rl}^x \mathcal{U}_{sl}^{x*} \quad (8)$$

This decomposition format is closely related to the existing low-rank approximations. For instance, it appears to have the same format as the tensor hypercontraction^{36–38} (THC) for CC amplitudes;³⁹ however, in our decomposition, the matrix $\mathcal{U}^x$ is restricted to be unitary so that the transformation can be efficiently realized by a orbital rotation, which is important in particular for quantum computation. It is also related to the low-rank decomposition of the CC amplitudes recently introduced by Motta and co-workers,²⁵ and originally by Peng and Kowalski⁴⁰ to 4-index two-electron repulsion integrals (ERIs), where the matrix $\mathcal{J}^x$ is replaced by a tensor product of two vectors, i.e., restricted to a rank-one matrix, consisting of singular values. Because $\mathcal{J}^x$ is of course allowed to be full-rank in our parametrizations, it is more flexible as will be shown for the ERIs in Appendix A. A low-rank decomposition scheme that was used for the comparison shown in Section 3.1 is given in Appendix B.

Instead of the single exponent form $e^{\sum_{x=1}^k \hat{\mathcal{T}}_{(x)}^{\text{uCJ}}}$, we adopt a product form of the uCJ exponential operators

$$e^{\hat{\mathcal{T}}_{(k)}^{\text{uCJ}}} \cdots e^{\hat{\mathcal{T}}_{(2)}^{\text{uCJ}}} e^{\hat{\mathcal{T}}_{(1)}^{\text{uCJ}}} = e^{-\hat{K}_k} e^{\hat{J}_k} e^{\hat{K}_k} \cdots e^{-\hat{K}_2} e^{\hat{J}_2} e^{\hat{K}_2} e^{-\hat{K}_1} e^{\hat{J}_1} e^{\hat{K}_1} \quad (9)$$

In this study, it is termed as k -uCJ ansatz since this splitting exponential operator form is analogous to the k -fold products of (k) -UpCCGSD ansatz recently introduced by Lee and co-workers,²⁶ in which a product of exponential operators of the pair coupled cluster doubles^{41–45} with the generalized singles. Note that the product form and the single-exponent form are not equivalent because the uCJ operators with different x are not commutative. In fact, by increasing k , the product form should make it possible to give the exact full-CI wave function, whereas the single-exponent form of the k -uCJ ansatz can merely reproduce the uCCGSD wave function, which is numerically very accurate but not exact. This conclusion on the exactness of the ansatz at the large k limit is deduced from the analysis given in the recent work by Evangelista.⁴⁶ Briefly, the uCCGSD operator can be decomposed to a product of the uCJ operator using the Trotter decomposition, and a product of the uCCGSD operators can reach an exact parametrization of the wave function as proved in ref 46.

2.3. Some Considerations for Implementation of the Exponential Ansatzs on Quantum Circuits. In principle, any unitary operation can be represented by quantum gates on universal quantum computer; however, so far, most of the quantum algorithms for quantum chemistry rely on the mapping of an exponential of a product of Pauli operators to a set of quantum gates consisting of two-qubit controlled gates and single-qubit gates.⁴⁷ Those Pauli operators are produced by the Jordan–Wigner or Bravyi–Kitaev transformations of the second-quantized operators of quantum chemistry.⁴⁸

For example, application of the elementary creation and annihilation operators are represented by the Pauli- X , Y , Z gates in the Jordan–Wigner transformation as

$$a_p^\dagger = \frac{1}{2}(\sigma_p^x \otimes \sigma_{p-1}^{z \rightarrow} - i\sigma_p^y \otimes \sigma_{p-1}^{z \rightarrow}) \quad (10)$$

$$a_p = \frac{1}{2}(\sigma_p^x \otimes \sigma_{p-1}^{z \rightarrow} + i\sigma_p^y \otimes \sigma_{p-1}^{z \rightarrow}) \quad (11)$$

where $\sigma_{p-1}^{z \rightarrow} \equiv \sigma_{p-1}^z \otimes \sigma_{p-2}^z \otimes \cdots \otimes \sigma_1^z$, and thus an exponential operator $e^{\hat{t}}$ is represented by an exponential of a summation of Pauli operator products, $e^{i\sum_l \theta_l (\prod_{\mu \in l} \sigma_\mu)}$. Note that the imaginary unit i in the exponent reflects unitarity of the cluster operator $\hat{t}$. To compute an exponential operator $e^{\hat{t}}$ on a quantum circuit accurately, one therefore needs to resort to the Trotter decomposition, e.g.,

$$e^{i\sum_l \theta_l (\prod_{\mu \in l} \sigma_\mu)} \simeq \left(\prod_l e^{i\theta_l n^{-1} \prod_{\mu \in l} \sigma_\mu} \right)^n \quad (12)$$

because the Pauli operator products $\prod_{\mu \in l} \sigma_\mu$ are usually not commutative to each other, which significantly increases the depth of the circuit. One way to avoid this problem is to simply adopt the single Trotter step form $\prod_l e^{i\theta_l \prod_{\mu \in l} \sigma_\mu}$, or n -step form, as an alternative ansatz. Hereafter, the Trotter n -step ansatz is denoted by a superscript “circ/ n ” to the corresponding original ansatz, e.g., the energy of the k -UpCCGSD^{circ/ n} ansatzs are evaluated by performing the n -times symmetric Trotter steps with the k -UpCCGSD ansatz, respectively. In fact, we found that not all exponential ansatz are virtually compatible with their Trotter n -step ansatzs as will be shown in Section 3.2, although it was demonstrated that the single Trotter step ansatz of uCCSD gave nearly identical variational energies to the original uCCSD ansatz for a H_2 molecule.⁴⁹

Importantly, the k -uCJ ansatz defined by eq 8 can be implemented without the Trotter approximation by nature. Because the operator $\hat{J}$ only have the number operators

$$a_p^\dagger a_p a_q^\dagger a_q = \frac{1}{4}(1 - \sigma_p^z)(1 - \sigma_q^z) \quad (13)$$

which are written by commutative σ^z matrices, and the orbital rotations $e^{\hat{K}}$ can be implemented using the Givens rotations,^{50,51} one can implement the k -uCJ to a quantum circuit without the use of Trotter approximation.

3. RESULTS AND DISCUSSION

An important field of applications of quantum computation in quantum chemistry is the multireference problem where rigorous algorithms with polynomial cost are not well established though a lot of effort has been devoted, such as ab initio density-matrix renormalization group theory.^{52,53} We examined the performance of various exponential ansatzs on the description of triple bonds dissociation of N_2 molecule with the STO-6G basis sets. The six Hartree–Fock canonical orbitals, HOMO – 2 to LUMO + 2, were used to construct the Fock space represented by a quantum register on a circuit simulator, i.e., the α and β spin-orbitals were assigned to 12 qubits. The ordering and characteristics of the six canonical orbitals remain unchanged between $r = 1.0$ and 2.4 Å bond length. We also examine the applicability of the k -uCJ ansatz for larger systems, hydrogen chains H_n up to $n = 10$.

3.1. Variational Minimization without the Trotter Approximation. First, we examined the accuracy of the k -uCJ ansatz and various existing exponential ansatzs with the HF reference function. At this point, the energy expectation values were evaluated not by a quantum circuit simulator but the standard matrix exponentiations, i.e., the Trotterization was completely absent. Figure 1a (top panel) shows the potential energy curves of the full-CI and k -uCJ ($k = 1, 2$, and 3) ansatzs and the SVD(k) approximation defined in Appendix B. The result of the conventional CCSD calculations is also shown for comparison.

The 1-uCJ works well at around the equilibrium bond length, e.g., the energy deviates from the full-CI by 6 mE_h at $r = 1.2$ Å where the multireference effect is not so significant, but it cannot describe the dissociation correctly. The error is rapidly decreased by increasing k , e.g., less than 1 mE_h at around the equilibrium bond length and at the dissociation. It was also observed that the convergence to the full-CI energy is relatively slow at the intermediate region and the maximum error is 7 mE_h with $k = 2$, and 3 mE_h with $k = 3$ at $r = 2.0$ Å.

The SVD(k) approximation was found to be problematic for this system. Not only the convergence of the deviation from the full-CI energy is slow even at the equilibrium bond length region, but we also found that it becomes pathologically slow when the multireference effect is important. In the first place, to describe the dissociation correctly with the SVD(k) approximation by increasing k , we needed to optimize the reference T_2 amplitude by more expensive ansatz that can correctly describe the dissociation as described in Appendix B.

Figure 1b (bottom panel) shows the potential energy curves obtained with k -uCJ and k -UpCCGSD ansatzs. The k -uCJ ansatz has similarity to the k -UpCCGSD ansatz since both take a product of the single-exponential operators uCJ or UpCCGSD, and the number of parameters grows only quadratically $O(N^2)$ with the number of orbitals N if k is

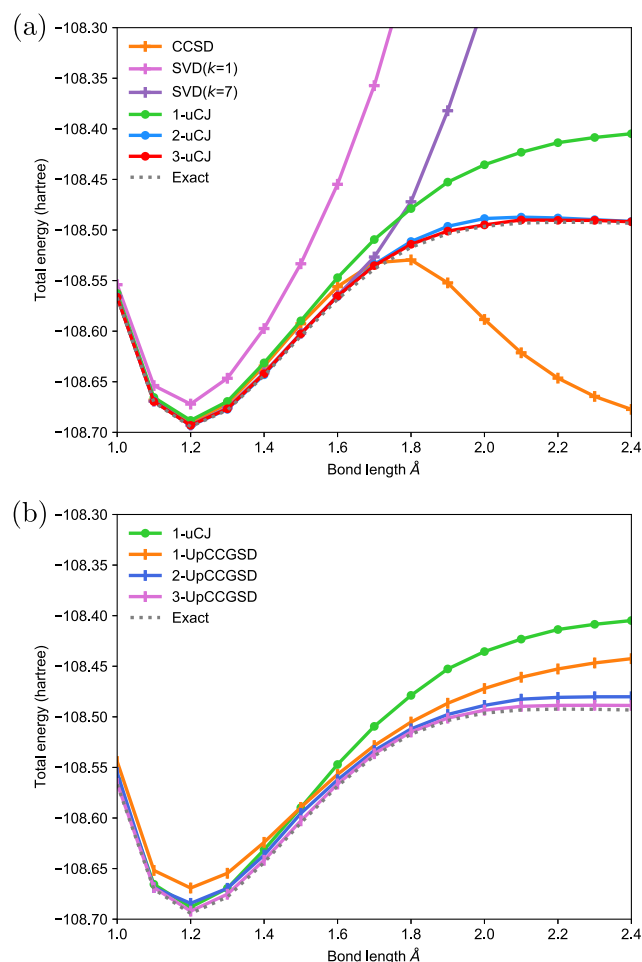


Figure 1. Potential energy curves for the N_2 molecule for the different ansatzes using the STO-6G basis set where the four lowest orbitals are frozen, i.e., six electrons are distributed to the other six orbitals. (a) SVD(k) and k -uCJ ($k = 1, 2$, and 3) ansatzes. (b) 1 -uCJ and k -UpCCGSD ($k = 1, 2$, and 3) ansatzes.

constant, and thus is expected to be expressed by a quantum circuit with linear $O(N)$ depth. Both 1 -UpCCGSD and 1 -uCJ ansatzes are obviously not adequate for describing the PEC, but the errors are rapidly decreased with increasing k , while the convergence to the full-CI energy is bit slower for the k -UpCCGSD than for the k -CJ in particular at around the equilibrium bond length and at the dissociation. All the energies were computed based on the original definitions written in the second quantized operator; thus, there is no guarantee that those can also be reproduced efficiently by the quantum gates on quantum circuits if the cluster operators are not commutative. This point will be examined in the next subsection.

3.2. Variational Energies with the Trotter Splitting Ansatz. In the previous subsection, all the energies were evaluated using matrix exponentiation as provided by the definitions written in the second quantization. As noted above, however, most of the exponential ansatzes, except for the k -uCJ, require the Trotter decomposition to be accurately computed on quantum circuits. Here, instead of performing the time-consuming many Trotter steps to reproduce the original ansatzes, we adopted the alternative ansatzes that are defined by the single or n -times Trotter steps for the summation over the Pauli operator products produced by the Jordan–Wigner

transformation of the original ansatzes. To be more specific, in the Trotter steps of the exponential operator, the anti-Hermite cluster operators $\hat{t}_{pq} = t_{pq} (a_{pa}^\dagger a_{qa} a_{pb}^\dagger a_{qb} - a_{qb}^\dagger a_{pb} a_{qa}^\dagger a_{pa})$ were arranged in row-major order where the canonical orbitals were numbered from the lowest energy.

Figure 2 shows the potential energy curves obtained with the 1 -UpCCGSD^{circ/ n} and k -uCJ^{circ/ n} ansatzes. As expected, the 2 -

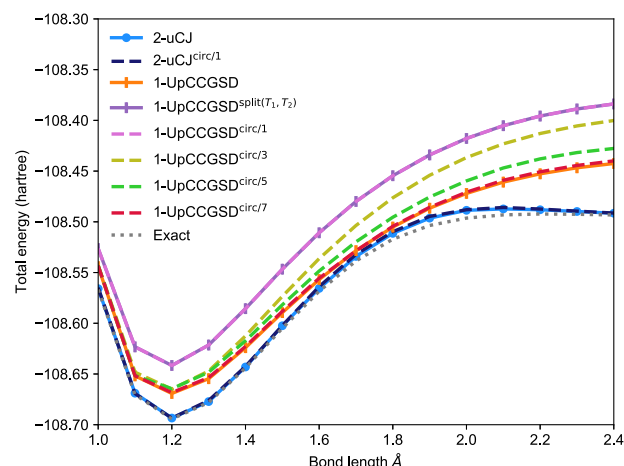


Figure 2. Potential energy curves for the N_2 molecule for the different ansatzes using the STO-6G basis set where the four lowest orbitals are frozen, i.e., six electrons are distributed to the other six orbitals. The superscript circ/ n denotes a variant of the original ansatz where the Pauli operator products are split in n -times Trotter steps.

u CJ^{circ/ 1} reproduced the results of the original 2 -uCJ ansatz because the Pauli operator products in the exponential operators are all commutative. However, it is somewhat surprising that the potential energy curve of the 1 -UpCCGSD^{circ/ 1} was much different from that of the original 1 -UpCCGSD ansatz; in fact, the 1 -UpCCGSD^{circ/ 1} is less powerful than the 1 -UpCCGSD ansatz. To find a clue to these discrepancies, we performed another variant of the 1 -UpCCGSD in which the singles and doubles operators were split as $e^{\hat{T}_1 - \hat{T}_1^\dagger} e^{\hat{T}_2 - \hat{T}_2^\dagger}$, instead of $e^{(\hat{T}_1 + \hat{T}_2) - (\hat{T}_1^\dagger + \hat{T}_2^\dagger)}$, hereafter denoted by 1 -UpCCGSD^{split- (T_1, T_2)} . Interestingly, the potential energy curve of the UpCCGSD^{split- (T_1, T_2)} is nearly identical to that of the 1 -UpCCGSD^{circ/ 1} . It clearly suggests that the large discrepancies between 1 -UpCCGSD and 1 -UpCCGSD^{circ/ 1} is mainly caused by splitting the singles and doubles cluster operators, which should be caused by splitting of the products of the Pauli operators in the single Trotter step approximation. As shown, 1 -UpCCGSD^{circ/ n} is converged to the result of the original 1 -UpCCGSD by increasing the number of the Trotter steps as $n = 1, 3, 5$, and 7 . It should be noted that the large discrepancy between the original form and a Trotter-splitting form of the k -UpCCGSD ansatzes is also discussed in more detail in ref 54, in which the magnitude of the error was shown to be correlated with the accuracy of the original ansatz, i.e., the error is magnified where the energy of the original k -UpCCGSD largely deviates from that of the full-CI, and the 1 -UpCCGSD shown here could be the worst case.

3.3. Benchmarks of the k -uCJ Ansatz for Hydrogen Chains. To examine the applicability of the k -uCJ ansatz for larger systems, benchmark calculations were performed for hydrogen chains H_n ($n = 4, 6, 8$, and 10) with the STO-3G basis sets, i.e., the full-CI space consists of distributing n

electrons in n orbitals. Table 1 shows the number of variational parameters used in each ansatz and the error in the correlation

Table 1. Performance of the k -uCJ Ansatz for the Linear Hydrogen Chain H_n ($n = 4, 6, 8$, and 10)^a

	H_4 (4e, 4o)	H_6 (6e, 6o)	H_8 (8e, 8o)	H_{10} (10e, 10o)
# of qubits	8	12	16	20
# of Variational Parameters				
1-uCJ	35	77	135	209
2-uCJ	70	154	270	418
FCI ^b	36	400	4900	63504
Percentage of Error in the Correlation Energies				
1-uCJ	0.85%	1.40%	1.90%	2.23%
2-uCJ	0.00%	0.05%	0.16%	
uCCSD	1.38%	2.58%	3.23%	
$E_{\text{corr}}^{\text{FCI}}$ (au)	-0.322164	-0.478771	-0.635260	-0.791736

^aThe distance between the nearest-neighbor atoms were all set to be 2.0 Å, where the ground state exhibits a significant multi-reference character. ^bNo. of determinants in the Hilbert space, $n_o C_{n_o} \times n_e C_{n_e}$, where n_o and n_e denotes the number of orbitals and $\alpha(\beta)$ electrons, respectively.

energy, i.e., $(E_{\text{corr}}^{k\text{-uCJ}} - E_{\text{corr}}^{\text{FCI}})/E_{\text{corr}}^{\text{FCI}}$ (%). The variational parameters are $\mathcal{J}^{\sigma\tau}$ and $\mathcal{K}$, of which the number of nonredundant parameters are $n(n+1)/2$ and $n^2 - 1$, respectively. In total, the number of variational parameters of the k -uCJ ansatz is equal to $(2n^2 + n - 1) \times k$. While the number of parameters of k -uCJ for the H_4 is comparable to or exceeds the size of the FCI space, and the correlation energy was fully recovered by the 2-uCJ, it becomes significantly less than the size of the space because it scales only quadratically with the number of orbitals. The distance between the nearest-neighbor hydrogen atoms were set to be 2.0 Å, where the ground states of hydrogen chains exhibit significant multi-reference character and capturing the electron correlation energy should be difficult. The error of k -uCJ slightly increased with n at least for up to H_{10} . The same is true for the uCCSD calculations, and the similar trend was also reported in ref 55. Further numerical analyses are needed to know if the error will converge to a moderate constant percentage.

4. SUMMARY

In this study, we present an efficient $O(N^2)$ -parameter ansatz for the VQE algorithm, named k -uCJ ansatz, which is derived as a tensor decomposition of the coupled cluster amplitudes, specifically generalized CC, and can reproduce the uCCGSD energy or rather will reach the exact full-CI energy because a sequential product form is adopted. Each term of the expansion is identical to the unitary variant of the cluster Jastrow (CJ) ansatz introduced by Neuscamman,³⁰ although the Jastrow parameters are restricted to pure imaginary by the unitarization and the physics captured by the ansatz should be different from that captured by the CJ ansatz (see Appendix C). It was confirmed that the k -CJ ansatz rapidly converged to the full-CI energy with increasing k , and the chemical accuracy can be obtained with $k = 3$ for the triple bond dissociation of the N_2 molecule.

The accuracy of the exponential ansatzs were examined not only with their original definition but also with their “hardware-friendly” variants, in which all the Pauli products

generated by the Jordan–Wigner transformation are split by a few, typically single, Trotter steps. Interestingly, the single Trotter step variant, UpCCGSD^{circ/1}, was found to be considerably less accurate than the original ansatz.

One of the advantages of the k -uCJ ansatz for quantum computation is that it does not need the Trotterization to be implemented on a quantum circuit because the cluster operators in each exponent are all commutative. The k -uCJ ansatz is therefore a good candidate for an efficient quantum computation with the near-term quantum computers. It should be noted that one drawback of the use of a product of exponential operator ansatz is that the parameter optimization is often trapped by the local minimum, as has been already reported by Lee,²⁶ we also encountered the same problem when the low-rank decomposition could not give a good initial guess. We also found that the use of the anti-symmetrized geminal power (AGP) reference function can improve the description of the k -uCJ, see Appendix C, although an efficient preparation of the AGP state on quantum computers is not trivial. An improvements of the reference function should be another issue to be explored.

In the NISQ era,²⁰ quantum computers can treat a few hundred orbitals at most, which is not quite large enough for describing the dynamical correlation in most cases. A straightforward approach to this problem is to adopt the active space model and use the VQE as an alternative to the full-CI in the active space, as done in the DMRG-CASSCF and subsequent dynamical correlation theories.^{56–63} In that case, the cumulant approximation to the high-order reduced density matrices should be necessary to reduce the required times of the measurements.^{64–66} Another approach is to cast the dynamical correlation into the active space Hamiltonian by similarity transformations in the first or second quantization, e.g., refs 67, 68 and 69–72, respectively, and many others, although it is not possible to list them all here. The k -uCJ ansatz is particularly suitable for this approach because the similarity transformed Hamiltonians usually possess more than two-body interactions and it is straightforward to extend the k -uCJ ansatz to more than two-body Jastrow factor forms.

5. COMPUTATIONAL DETAILS

An in-house code written in Python 3 was used to perform all the calculations. The HF canonical orbitals and molecular integrals were generated by the PySCF library,⁷³ and the fermionic algebra for the exponential ansatz in the Fock space and the mapping to a qubit representation by the Jordan–Wigner transformation were handled using the OpenFermion library.⁷⁴ The quantum circuit simulations were performed by the Qulacs library.⁷⁵ The low-rank decomposition of the conventional CCSD amplitudes was adopted to generate an initial guess of the variational optimization for the k -uCJ ansatz. It works well in many cases, but otherwise the optimization was repeatedly performed starting from randomly chosen points until a smooth PEC was obtained.

■ APPENDIX A: JASTROW-TYPE DECOMPOSITION OF THE HAMILTONIAN

The four-index two-electron repulsion integrals (ERIs) in ab initio Hamiltonian, $\hat{H} = \sum_{pq, \sigma\tau} f_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \sum_{pqrs, \sigma\tau} h_{pqrs} a_{p\sigma}^\dagger a_{q\sigma}^\dagger a_{r\tau} a_{s\tau}$, can also be decomposed by the Jastrow-type parametrization, which is rather simpler form than that for T_2 amplitudes in the Section 2.2 due to its 8-fold symmetry

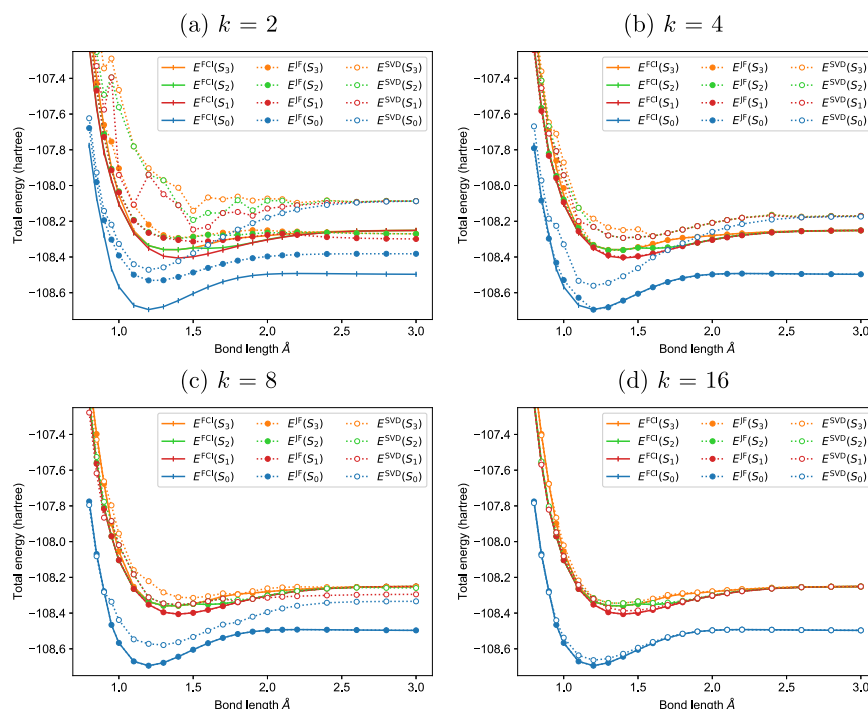


Figure 3. Potential energy curves for the low-lying singlet states of the N_2 molecule using the STO-6G basis set where the four lowest orbitals are frozen, i.e., six electrons are distributed to the other six orbitals. E^{IF} and E^{SVD} were obtained by diagonalizing the approximated Hamiltonian given by eqs 14 and 16, respectively.

$$h_{pq,rs} \simeq h_{pq,rs}^{\text{IF}} = \sum_x \sum_{jl} U_{pj}^x U_{qj}^x U_{rl}^x U_{sl}^x \quad (14)$$

$$\begin{aligned} \hat{H} &\simeq \hat{H}^{\text{IF}} = \sum_{pq,\sigma} f_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \sum_{x=1}^k \sum_{jl,\sigma\tau} J_{jl}^x \tilde{a}_{j\sigma}^{x\dagger} \tilde{a}_{j\sigma}^x \tilde{a}_{l\tau}^{x\dagger} \tilde{a}_{l\tau}^x \\ &= \sum_{pq,\sigma} f_{pq} a_{p\sigma}^\dagger a_{q\sigma} + \sum_{x=1}^k e^{-\hat{K}^x} \hat{J}^x e^{\hat{K}^x} \end{aligned} \quad (15)$$

where $\tilde{a}_{j\sigma}^{x\dagger} = \sum_p a_{p\sigma}^\dagger U_{pj}^x$ and K^x and J^x are real matrices as in eqs 3 and 4. This Jastrow-type parametrization should be compared with the low-rank approximation to Hamiltonian introduced by Peng and Kowalski⁴⁰ and recently by Motta and co-workers.^{5,25}

$$\begin{aligned} h_{pq,rs} &\simeq h_{pq,rs}^{\text{SVD}} = \sum_{x=1}^k \left(\sum_j U_{pj}^x \lambda_j^x U_{qj}^x \right) \left(\sum_l U_{rl}^x \lambda_l^x U_{sl}^x \right) \\ &= \sum_x \sum_{jl} U_{pj}^x U_{qj}^x \lambda_j^x \lambda_l^x U_{rl}^x U_{sl}^x \end{aligned} \quad (16)$$

While full-rank matrices J^x are not separable for the indices j and l and thus may not be suitable for reducing the cost of the atomic-orbital integral transformation in classical algorithms, the Jastrow-type parametrization is again more flexible than the low-rank approximation with the same number of terms, k , as shown in Figure 3, in which the parameters J^x and K^x were determined by minimizing the error $\Delta = |\mathbf{h} - \mathbf{h}^{\text{IF}}|$. The Hamiltonian in eq 15 is expressed by the orbital rotations $e^{\hat{K}^x}$ and the product of number operators $\hat{n}_{j\sigma} \hat{n}_{l\tau}$, which scales only quadratically with the number of orbitals. This should serve to reduce the cost of a real-time propagation with $e^{-i\hat{H}t}$ or the sampling for energy expectation values in VQE on quantum computers, e.g., ref 76 and references therein.

■ APPENDIX B: LOW-RANK APPROXIMATION TO THE GENERALIZED UCC AMPLITUDES

The uCCG doubles cluster operator can be rewritten as

$$\begin{aligned} \hat{t}_2^{\text{uCCG}} &= \hat{T}_2^{\text{CCG}} - \hat{T}_2^{\text{CCG}\dagger} = \sum_{pqrs} \tau_{pqrs}^{\text{uCCG}} \hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_r \hat{a}_s \\ &= \frac{1}{2} \sum_x \left\{ \left(\sum_{pq} \mathcal{L}_{pq}^{x+} \hat{a}_p^\dagger \hat{a}_q \right)^2 + \left(\sum_{pq} \mathcal{L}_{pq}^{x-} \hat{a}_p^\dagger \hat{a}_q \right)^2 \right\} \\ &= \frac{1}{2} \sum_x \left\{ \left(\sum_{pq} \left(\sum_j \mathcal{U}_{pj}^{x+} \lambda_j^{x+} \mathcal{U}_{qj}^{x+*} \right) \hat{a}_p^\dagger \hat{a}_q \right)^2 \right. \\ &\quad \left. + \left(\sum_{pq} \left(\sum_j \mathcal{U}_{pj}^{x-} \lambda_j^{x-} \mathcal{U}_{qj}^{x-*} \right) \hat{a}_p^\dagger \hat{a}_q \right)^2 \right\} \end{aligned} \quad (17)$$

where $\mathcal{L}^\pm$, $\mathcal{U}^\pm$, and $\lambda^\pm$ are obtained by applying SVD to T^{CCG}

$$\begin{aligned} T_{pq,rs}^{\text{CCG}} &= \sum_x V_{pq,x} \sigma_x V_{rs,x} \\ &= \sum_x (\sqrt{\sigma_x} V_{pq,x}) (\sqrt{\sigma_x} V_{rs,x}) \\ &= \sum_x \frac{\sigma_x}{|\sigma_x|} I_{pq}^x I_{rs}^x \end{aligned} \quad (18)$$

and applying SVD again to $\mathcal{L}^{x\pm}$

$$\begin{aligned} \mathcal{L}_{pq}^{x\pm} &= \sqrt{\frac{\sigma_x}{|\sigma_x|}} (L_{pq}^x \pm iL_{qp}^x) \quad (19) \\ &= \sum_j \mathcal{U}_{pj}^{x\pm} \lambda_j^{x\pm} \mathcal{U}_{qj}^{x\pm*} \quad (20) \end{aligned} \quad (19a)$$

Hence, the amplitude results in

$$\begin{aligned} \tau_{pqrs}^{\text{uCCG}} &= \frac{1}{2} \sum_x \sum_{jl} \{ \mathcal{U}_{pj}^{x+} \mathcal{U}_{qj}^{x+*} (\lambda_j^{x+} \lambda_l^{x+}) \mathcal{U}_{rl}^{x+} \mathcal{U}_{sl}^{x+*} \\ &\quad + \mathcal{U}_{pj}^{x-} \mathcal{U}_{qj}^{x-*} (\lambda_j^{x-} \lambda_l^{x-}) \mathcal{U}_{rl}^{x-} \mathcal{U}_{sl}^{x-*} \} \quad (21) \end{aligned}$$

Note that because L^x are not generally symmetric matrices, the transformation by eq 19a is needed to make $\mathcal{L}^{x\pm}$ normal matrices and to ensure the existence of unitary matrices $\mathcal{U}^{x\pm}$.

In the low-rank approximation denoted by SVD(k), the summation over x in eq 21, where the singular values σ_x are assumed to be sorted in descending order of $|\sigma_x|$, was truncated at $M = k$, while no truncation was made for another SVD in eq 19a. Because the reference T^{CCG} in eq 18 is generally unknown a priori, we optimized with the uCCGD ansatz and performed the decomposition for the SVD(k) calculations in Section 2.2, although the uCCGD calculations were more expensive than the energy evaluation with the SVD(k) approximation. The conventional CCSD amplitudes were used in ref 25 as an alternative; however, it would not be appropriate as a reference when the multireference effect is significant. While the amplitudes obtained by the SVD(k) can be reproduced by setting $\mathcal{J}_{jl}^x = \lambda_j^{x+} \lambda_l^{x+}$, $\mathcal{U}_{pj}^x = \mathcal{U}_{pj}^{x+}$ in eq 8 and adding its complex conjugate, it should be much more flexible than eq 21 because $\mathcal{J}^x$ are not restricted to the rank-one matrices.

■ APPENDIX C: UCJ-AGP ANSATZ

The CJ exponential operator was originally combined with the anti-symmetric geminal power (AGP) reference wave function in the original CJ-AGP ansatz,³⁰ which are expected to serve complementary roles to describe the electron correlation and can give a better description than that with the HF reference. In particular, the Hilbert space Jastrow factor, namely, the CJ, can completely fix the size-consistency error of the AGP wave functions,³⁴ which was motivated by the work of Sorella^{77,78} on the real-space Jastrow factor, because it can act as a particle number projector to penalize the configurations with incorrect particle distributions of the AGP, which is not the case of its unitary variant, the uCJ operator.

Figure 4 (bottom panel) shows the potential energy curves obtained with k -uCJ and uCJ-AGP ansatzs. The result of 1-uCJ with AGP ansatz, in which the exponential operator of 1-uCJ was applied to the AGP reference wave function and all the parameters including the AGP parameterized by

$$|\Psi^{\text{AGP}}\rangle = \left(\sum_{pq} F_{pq} \hat{a}_{p\alpha}^\dagger \hat{a}_{q\beta}^\dagger \right)^N |\text{vac}\rangle \quad (22)$$

was simultaneously optimized. It was found that the unitary variant, 1-uCJ-AGP ansatz, well reproduced the full-CI results both at equilibrium bond length region and at dissociation region with less than 5 mE_h error. It should be noted that a relatively large error was observed at the intermediate region, e.g., ~25 mE_h error at $r = 1.7$ Å, although the CJ-AGP ansatz almost perfectly reproduced the full-CI result for whole the

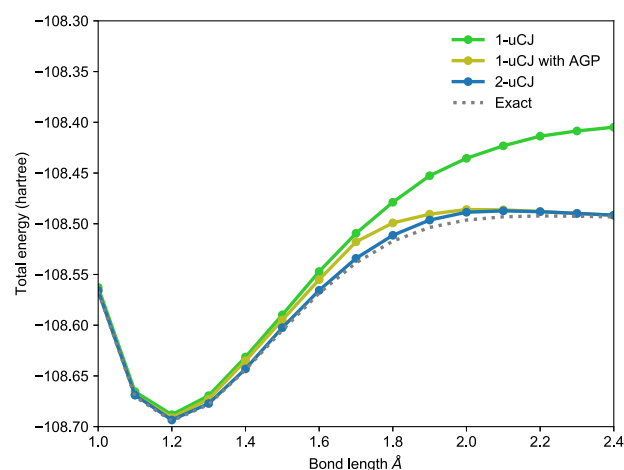


Figure 4. Potential energy curves for the N₂ molecule for the different ansatzs using the STO-6G basis set where the four lowest orbitals are frozen, i.e., six electrons are distributed to the other six orbitals.

bond stretching ($r = 1.0$ – 1.8) in the original work by Neuscamman.

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Notes

The authors declare no competing financial interest.

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